CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2011

CL 101 Fundamentals of civil engineering

Date: 13.12.2011, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 **(a)** Define hydrology. **[01]**

(b) Enlist branches of civil engineering & explain any two in detail. **[06]**

Q - 2 **(a)** Draw the conventional sign to show the brick work and R.C.C. work. **[02]**

(b) Write a note on types of residential building. **[03]**

(c) Discuss any two principles of building planning in detail. **[04]**

(d) Explain the hydrological cycle with neat sketch. **[05]**

OR

Q - 2 **(a)** Define hydraulic structure. Give classification of dam. **[02]**

(b) Briefly describe the methods of conservation of water. **[03]**

(c) Give the functions of Window, Foundation, Stair and Beam. **[04]**

(d) Classify the building based on occupancy. **[05]**

Q - 3 **(a)** Define super structure and sub-soil. **[02]**

(b) Discuss in detail frame structure. **[03]**

(c) Explain set-back and light plane. **[04]**

(d) Write a note on sources of water. **[05]**

OR

Q - 3 **(a)** Enlist structural components and functional components. **[02]**

(b) Discuss canal classification. **[05]**

(c) Draw the detailed section at A-A' for the line diagram shown in **figure 1.** **[07]**

Take scale: 1 cm = 0.5 m

SECTION – II

Q - 4 (a) Bearing of lines of a closed traverse are as given in table below: [07]

LINE	FORE BEARING
AB	N 45^O E
BC	N 75^O E
CD	S 35^O W
DA	N 65^O W

Find the included angles and apply necessary checks.

Q - 5 (a) The length of a chain line when measured with a 20 m chain was found to be 1432 m. when a 30 m chain was 0.65 m too short was used for the purpose , the line was found to be 1445m long. Find the error in 20 m chain. [05]

(b) Write classification of surveying. [05]

(c) Explain the terms Magnetic declination, Local attraction, Bench mark, Tie station [04]

OR

Q - 5 (a) Sketch symbols of Chain line and Main station [02]

(b) Write the reduced bearing of 335° and 240° [02]

(c) The following readings are taken on continuously falling ground with staff of 4.0 m. They are 0.400, 0.765, 1.270, 2.560, 3.220, 3.950, 0.390, 1.690, 3.500, 0.800, 1.920, 2.450 and 3.980. The instrument was shifted after 6th and 9th reading. Enter the reading in the page of level book and calculate the RLs of all the points by Height of Instrument method. If the first reading was taken on benchmark of 100.000 m. [10]

Q - 6 (a) Discuss role of transportation in national development. [07]

(b) Differentiate between plane surveying and geodetic surveying. . [04]

(c) Explain the Plan and Map terms in detail. [03]

OR

Q - 6 (a) Enlist various traffic control devices. [03]

(b) Write advantages and disadvantages of railway. [04]

(c) Write a note on various types of compass. [07]

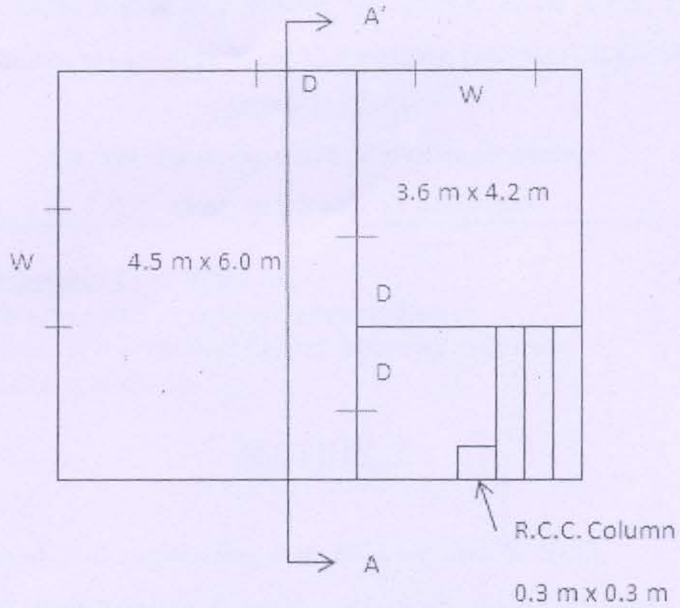

Figure : 1

Sr. No.	Components	Size
1.	D	1.2 m x 2.1 m
2.	W	1.2 m x 1.4 m
3.	Tread	20 cm
4.	Riser	15 cm

The height of room is 3.1 m.

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